# Indian Institute of Engineering Science and Technology, Shibpur

B. Tech (8th) Semester Mid-Semester Examinations, 2025

Entrepreneurship and Innovation Management (MS4262)

Full Marks: 30 | Time: 2 hours

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**(Group – A)**

**Answer any 5 questions:** **[5 x 4 = 20]**

Q1: In your own words, state what is meant by the term innovation? What are the major types of innovation? Give examples.

Q2: How can marketing research help an entrepreneur identify a market? Which is of greater value to the entrepreneur, primary or secondary data? Why?

Q3: Is there any difference between small business owner and entrepreneur? How?

Q4: Explain the entrepreneurial strategy matrix and recommend appropriate strategies.

Q5: Illustrate with a diagram showing a venture development stage.

Q6: Firms that fail to innovate will die. What does this statement mean in the context of new ventures?

**(Group – B)**

**Answer this question: (Compulsory)** **[10]**

Q7: "Entrepreneurship as a Career"- comment on this statement. What are the factors favouring entrepreneurship as a career option?

**Indian Institute of Engineering Science and Technology, Shibpur**

CST 8th Semester B Tech Midterm Examination, February 2025

**Quantum Computing (CS-4226)**

Full Marks 30 Time 120 minutes

Answer any five questions $((3+3) \times 5 = 30)$

OR

Answer any ten out of the fourteen part questions $(3 \times 10 = 30)$

1. (a) Explain the operation of Hadamard gate and show its Reversibility property.
   (b) Which gate is universal gate in quantum computing? Realize at least two common classical gates (XOR, OR, AND, NAND) using such gate.

2. The Pauli matrices are $X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ $Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$ $Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$
   (a) Find the adjoint of these matrices and explain the significance.
   (b) What role is played by the Pauli matrices in realization of superdense coding?

3. (a) Define information entropy for two bit classical binary system and show how information entropy changes with respect to the probability $p$ of symbol 0.
   (b) What is the long term probability of 0 and 1 for Markov source where $P(0|1) = p$ and $P(1|0) = q$?

4. Write short notes around the following statements:
   (a) Copying or cloning of qubits is not possible
   (b) Quantum teleportation looks exciting but is not realistic

5. (a) What is channel capacity and how does it affect the message rate for error-free communication?
   (b) Find a relation between the message rate and error probability for repetitive codes.

6. (a) Describe the fast Fourier transform algorithm (FFT) in classical computing domain.
   (b) Obtain the time complexity of classical FFT algorithm and argue whether improvement is needed.

7. (a) Briefly explain how quantum parallelism works.
   (b) Show how Deutsch algorithm exploits the idea of quantum parallelism.

# Indian Institute of Engineering Science and Technology, Shibpur

**B. Tech 8th Semester Mid-Semester Examination,**

**February 2025**

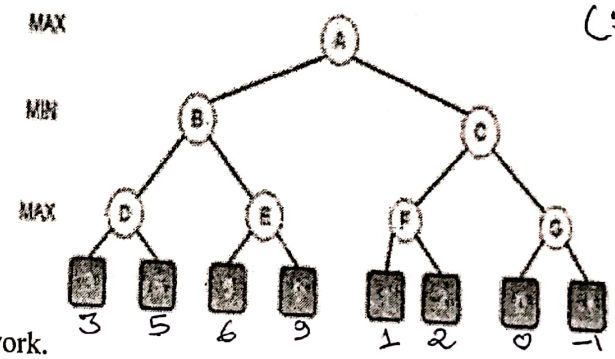
**Artificial Intelligence (CS4201)**

**Full Marks: 30** **Time: 2 Hours**

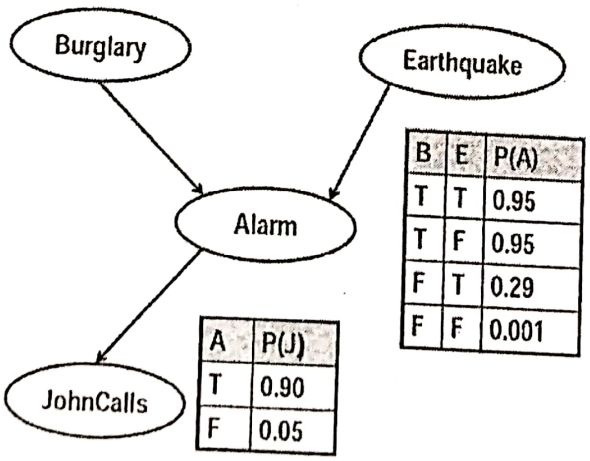
1. (i) How epsilon-greedy algorithm handles exploration exploitation dilemma in Reinforcement Learning Algorithm?
   (ii) What is the role of $\gamma$ function to calculate reward in Reinforcement Learning? 6+4

2. (i) What are the drawbacks of Steepest Ascent Hill Climbing Problem?
   (ii) What are the solutions? 5+5

3. (i) How many number of nodes are generated in a Depth-limited search and Iterative deepening search considering depth $d = 4$ and branching factor $b = 12$?
   (ii) Draw the game tree after applying alpha-beta pruning algorithm on the following tree. (5+5)




4.(i) Define Bayesian Belief Network.
(ii) Find out joint probability distribution of the following events considering the following network.
(a) Alarm has sounded but neither burglary, nor an earthquake occurred, however John calls Harry. (b) Alarm has sounded and there is burglary. 3+(4+3)




| B | E | P(A) |
|---|---|---|
| T | T | 0.95 |
| T | F | 0.95 |
| F | T | 0.29 |
| F | F | 0.001 |

| A | P(J) |
|---|---|
| T | 0.90 |
| F | 0.05 |